WS-01: Test Card TC-01 — Ground Bench Test Worksheet

STATIC THRUST, ELECTRICAL POWER & THERMAL EQUILIBRIUM (RIGID LOAD STAND)

AIRCRAFT: MCO-1D Mark 1	**CARD ID:** TC-01	**DATE:** ____ / ____ / 2026	**LOCATION:** _____
OPERATOR / TECH: _____	**AIRFRAME S/N:** MCO-1D-001	**AMBIENT TEMP:** _____ °C	**BARO / QNH:** _____ hPa

1. Flapping Frequency Sweep & Static Performance Matrix

Mount airframe securely to rigid load stand. Execute frequency steps sequentially. Record peak readings during final 10s of dwell.

Pt	Cmd Freq	Stroke Amp	Dwell	Target Thrust	Measured Thrust	GS Bus Current	Peak Motor Temp	ODrive FET Temp	Cable Slip OK?
P1	1.0 Hz	± 60.0°	30 s	*≥ 8.0 N*	[ ] N	[ ] A	[ ] °C	[ ] °C	[ ] OK
P2	1.5 Hz	± 70.0°	30 s	*≥ 16.0 N*	[ ] N	[ ] A	[ ] °C	[ ] °C	[ ] OK
P3	2.0 Hz	± 80.0°	30 s	*≥ 24.0 N*	[ ] N	[ ] A	[ ] °C	[ ] °C	[ ] OK
P4	2.5 Hz	± 80.0°	45 s	*≥ 31.0 N*	[ ] N	[ ] A	[ ] °C	[ ] °C	[ ] OK
P5	3.0 Hz	± 80.0°	60 s	*≥ 37.0 N*	[ ] N	[ ] A	[ ] °C	[ ] °C	[ ] OK
P6	3.5 Hz	± 70.0°	20 s	*≥ 42.0 N*	[ ] N	[ ] A	[ ] °C	[ ] °C	[ ] OK

2. Electrical Subsystem, Servo Rail & Structural Dynamics Checks

Check / Parameter Description	Sensor / Measuring Point	Target Pass Criteria	Measured Value	Evaluation
7.4V Servo Bus Rail Loaded Voltage	Oscilloscope across 7.4V rail at P5	*≥ 7.10 V (droop ≤ 0.3 V)*	[ ] V	[ ] PASS [ ] FAIL
16-Servo Simultaneous Reversal Surge	Current clamp on 2S battery lead	*≤ 45.0 A peak transient*	[ ] A	[ ] PASS [ ] FAIL
6S Main Bus Voltage at Full 3.5 Hz Load	ODrive vbus telemetry	*≥ 21.60 V (no severe sag)*	[ ] V	[ ] PASS [ ] FAIL
Dyneema Max Angular Slippage	Witness mark alignment on sector rim	*≤ 0.5° (zero visible shift)*	[ ] °	[ ] PASS [ ] FAIL
Airframe RMS Accelerometer Vibration	Pixhawk 4 Mini IMU log (10–200 Hz)	*≤ 2.0 g RMS*	[ ] g	[ ] PASS [ ] FAIL
Wing Spar Deflection at Root under P6	Optical deflection scale at station yoke	*No delamination, crack, or slip*	[ ]	[ ] PASS [ ] FAIL

GATE CERTIFICATION: TC-01 GROUND BENCH TEST QUALIFICATION	**STATUS: [ ] PASS [ ] CONDITIONAL [ ] FAIL / RE-TEST**
INSPECTOR / LEAD: _____	**SIGNATURE & DATE:** _____
REMARKS / DISCREPANCIES: _____	